

Analysis of the M/E/I Framework

Three Critical Points and Their Verification

This document was prepared by the author (J. Pascher) using AI assistance. The verification of all calculations through Python, the identification of the unit error, and the methodological objections (Λ CDM circularity, assumptions table) were formulated by the author.

Important Correction: An initial evaluation contained a unit error: $k_B = 1.38 \times 10^{-16}$ erg/K was used as photon energy instead of $k_B T = 1.38 \times 10^{-12}$ erg at $T = 10,000$ K (factor 10^4 forgotten). Correct rate: ≈ 0.004 photons/atom/year. Discrepancy factor: $\approx 64,000\times$.

Contents

1. Background: The M/E/I Framework
 2. All Assumptions Listed
 3. Python Verification
 4. Point 1: Bremsstrahlung as Landauer Erasure
 5. Point 2: Photon Rate from First Principles
 6. Dependence on Λ CDM
 7. Point 3: Prediction Without Fitting
 8. Numerical Check: Damiano's Arithmetic
- Overall Assessment · References

1. Background: The M/E/I Framework

Damiano Viglione applies Vopson's mass-energy-information equivalence (M/E/I) to the dark matter problem. The central equation is Landauer's principle in relativistic form:

m = (k2 · T · ln 2) / c^2

Each Bremsstrahlung photon represents an information erasure with effective mass. The historical radiation entropy leaves a gravitational memory effect.

2. All Assumptions Listed

Symbol legend: ✓ = physically justified · △ = plausible but not rigorously derived · ✗ = not derived / wrong

No.	Assumption	Status
A1	Landauer's principle applies to Bremsstrahlung	△ Requires new physics
A2	Each emission = erasure of exactly 1 bit	△ Not derived
A3	250 photons/atom/year	✗ First principles: ≈0.004
A4	10% of HI gas in starburst	△ No derivation
A5	Starburst duration: 5×10 ⁸ years	✓ Astrophysically plausible
A6	Plasma temperature: T = 10,000 K	✓ Typical for H-II regions
A7	Information remembers gravitationally without decay	✗ Not derived
A8	Linear scaling of all atoms	△ Spatial distribution ignored
A9	m = k ₂ T·ln2/c ² in astrophysical plasmas	△ Not confirmed
A10	No dispersal of information-mass	✗ Radiation leaves the system

Critical assumptions: A1, A2, A3, A7, A10. A3 is directly quantifiable: factor ≈ 64,000 too large.

3. Python Verification: Key Numbers

```
kB=1.380649e-23; c=2.99792458e8; ln2=0.693147; yr_s=3.1557e7 M_sun=1.989e30 # Bit weight at T=10,000 K m_bit = kB*1e4*ln2/c**2 # = 1.0648e-36 kg [Damiano: 1.06e-36, dev. 0.45%] # Bremsstrahlung (Rybicki & Lightman 1979, n_e=n_i=1e3, T=1e4 K, g=1.2) P_vol = 1.42e-27*(1e4)**0.5*1e3*1e3*1.2 # 1.704e-19 erg/cm3/s P_atom = P_vol/1e3 # 1.704e-22 erg/s kBT_erg = kB*1e4*1e7 # 1.381e-12 erg [CORRECT] N_dot = P_atom/kBT_erg # 1.234e-10 photons/s N_yr = N_dot*yr_s # ~0.004 photons/atom/year # M/E/I: 250 => discrepancy factor ~64,000
```

4. Point 1: Bremsstrahlung as Landauer Erasure

Possible in principle — technically demanding.

During Bremsstrahlung: $e^- + \text{Ion} \rightarrow e^- + \text{Ion} + \gamma$. The mapping $p_{\text{before}} \rightarrow p_{\text{after}}$ is *not uniquely reversible*.

M-value: How many predecessor states?

From the consistency condition (photon energy = Landauer energy):

$$E_{\gamma} = \log_2(M) \cdot k_2T \cdot \ln 2 = k_2T \Rightarrow \ln(M) = 1 \Rightarrow M = e \approx 2.718$$

On average e predecessor states lead to one final state. Mathematically consistent, but no rigorous QED derivation.

5. Point 2: Photon Rate from First Principles

Total power per volume (Rybicki & Lightman 1979):

$$P_{\text{vol}} \approx 1.42 \times 10^{-27} \cdot T^{\frac{1}{2}} \cdot n_e \cdot n_i \cdot g = 1.704 \times 10^{-19} \text{ erg/cm}^3/\text{s}$$

Power per atom:

$$P_{\text{atom}} = 1.704 \times 10^{-22} \text{ erg/s}$$

Photon energy at $T = 10^4 \text{ K}$ (correct):

$$h\nu \approx k_2T = 1.381 \times 10^{-23} \times 10^4 \times 10^7 = 1.381 \times 10^{-12} \text{ erg}$$

Unit error: An initial evaluation used $k_2T \approx 1.38 \times 10^{-16} \text{ erg}$ ($= k_2$ in erg/K, without multiplication by T). Correct: $1.38 \times 10^{-12} \text{ erg}$. Error: factor 10^4 .

Photon rate per year:

$$\dot{N}_{\gamma} = P_{\text{atom}} / (k_2T) \times t_{\text{year}} \approx 0.004 \text{ photons/atom/year}$$

Quantity	M/E/I Framework	First Principles
Photon rate	250 / atom / year	≈ 0.004 / atom / year
Discrepancy	—	Factor $\approx 64,000$

Consequence for DDO 154:

$$M_{\text{info(correct)}} = (0.004/250) \times 1.98 \times 10^9 \text{ M}_{\odot} \approx 31,700 \text{ M}_{\odot} \text{ [Target: 2–3 billion M}_{\odot}]$$

Discrepancy cannot be closed by parameter adjustment

For n_e alone: $n_e \approx 10^3 \times 64,000 = 6 \times 10^7 \text{ cm}^{-3}$ — **1,000× denser than Orion Nebula** ($\sim 10^4 \text{ cm}^{-3}$). Higher temperature makes it worse ($\dot{N} \propto T^{-1/2}$). Other processes: factor 2–10 at best.

6. Dependence on Λ CDM

Quantity	Source	Λ CDM-dependent?
Bremsstrahlung formula	QED / plasma physics	✓ No

Quantity	Source	Λ CDM-dependent?
n_e, T	H-II observations	✓ No
HI gas mass	SPARC, 21cm line	✓ No
Rotation curves	Doppler measurement	✓ No
Required DM mass (2–3 billion $M_\odot$)	NFW halo fit	✗ Yes — Λ CDM output
Starburst duration	Galaxy evolution models	△ Partially

Circularity: The target (2–3 billion $M_\odot$) comes from an Λ CDM NFW halo fit. M/E/I reproduces an Λ CDM output using non- Λ CDM physics. If Λ CDM is wrong, the target is also wrong. The clean test: reproduce the rotation curve directly — without Λ CDM as intermediate step.

7. Point 3: Prediction Without Fitting

Possible in principle — standard methodology. Model structure:

$$\rho_{\text{info}}(r) = \beta \cdot \dot{N}_\gamma(r)$$

$$v(r) = \sqrt{[G/r \cdot \int_0^r 4\pi r'^2 (\rho_{\text{bar}} + \rho_{\text{info}}) dr']}$$

Fixed β from other galaxies, publish $v(r)$ **before consulting observational data**.

8. Numerical Check: Damiano's Arithmetic

DDO 154: $N_{\text{bits}} = 2.97 \times 10^{64} \times 250 \times 5 \times 10^8 = 3.7125 \times 10^{75} \checkmark M_\odot = 1.987 \times 10^9 \checkmark$ (deviation 0.45%) **IC 2574:** $M_\odot = 8.231 \times 10^9 \checkmark$ (deviation 0.50%) **The arithmetic is internally correct. The problem lies exclusively in assumption A3 (250).**

Overall Assessment

Point	Possible?	M/E/I fulfilled?
1. Bremsstrahlung = Landauer	Yes	No
2. Photon rate first principles	Yes	No ($0.004 \neq 250$)
3. Prediction without fitting	Yes	No

The number 250 is $\approx$ 4–5 orders of magnitude too large. Damiano's arithmetic is internally correct (deviation 0.45%). The problem is exclusively assumption A3. The direction — dark matter through historical radiation entropy — deserves a rigorous foundation.

FFGFT connection: The T-field ($T \cdot m = 1$) explains dark matter effects geometrically from T^4 — without information-mass.

References

- Landauer, R. (1961). Irreversibility and heat generation. *IBM Journal*, 5, 183–191.
- Vopson, M. M. (2019). Mass-energy-information equivalence. *AIP Advances*, 9, 095206.
- Rybicki, G. B. & Lightman, A. P. (1979). *Radiative Processes in Astrophysics*. Wiley.
- Vaccaro, J. A. & Barnett, S. M. (2011). Information erasure without energy cost. *Proc. R. Soc. A*.
- Bérut, A. et al. (2012). Experimental verification of Landauer's principle. *Nature*, 483, 187.